



GIRRAWEE HIGH SCHOOL
TASK 1 2021 (*December 2021*)

MATHEMATICS
EXTENSION 1

Time allowed – 90 Minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be marked clearly Question 6, Question 7 etc. Each question is to be answered on a separate page in your answer booklet.
- You may ask for spare answer booklets if you need them.
- For Multiple choice: Fill in the circle corresponding to the correct answer on the multiple choice answer sheet in your answer booklet.

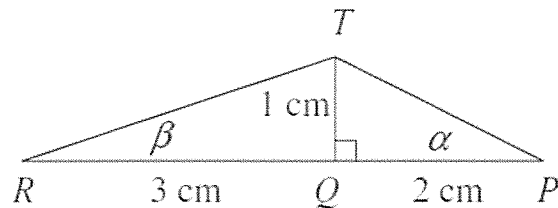
Question 6 (12 marks)

a) Show that

i) $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$ 3

ii) Hence, simplify $\sin^2 75^\circ - \sin^2 15^\circ$ 1

b)



i) Use the diagram above diagram to find the exact value for $\sin(\alpha + \beta)$. 3

ii) Hence show that the angles α and β in the diagram above have sum 45° 1

c) Use the compound -angle and double -angle results to find the exact value of :

i) $\frac{\tan 110^\circ + \tan 25^\circ}{1 - \tan 110^\circ \tan 25^\circ}$ 2

ii) $\sin \frac{8\pi}{9} \cos \frac{2\pi}{9} - \cos \frac{8\pi}{9} \sin \frac{2\pi}{9}$ 2

Question 7(11 marks)

- a) Use $t = \tan \frac{\theta}{2}$ to find the exact value of $\frac{1 - \tan^2 \frac{3\pi}{8}}{1 + \tan^2 \frac{3\pi}{8}}$ 2
- b) Prove that $\cos \theta \left(\tan \theta - \tan \frac{\theta}{2} \right) = \tan \frac{\theta}{2}$, using $t = \tan \frac{\theta}{2}$. 3
- c) i) If $t = \tan 75^\circ$, show that $\frac{2t}{1-t^2} = -\frac{1}{\sqrt{3}}$ 2
- ii) Hence show that $\tan 75^\circ = 2 + \sqrt{3}$ 1
- d) Solve $7 \sin x - 4 \cos x = 4$, for $0^\circ \leq x \leq 360^\circ$, by using the substitution $t = \tan \frac{1}{2} x$. 3

Question 8(13 marks)

Use the products– to– sums identities to show that

- a) i) $\cos 36^\circ - \cos 84^\circ = \sqrt{3} \sin 24^\circ$ 2
- ii) $\frac{\sin 3\theta + \sin \theta}{\cos 3\theta + \cos \theta} = \tan 2\theta$ 3
- b) Solve the following equations for $0 \leq x \leq 2\pi$
- i) $\sin 2x + \sqrt{3} \cos x = 0$ 3
- c) i) Express $\sqrt{3} \cos x + \sin x$ in the form $A \cos(x - \theta)$ where $A > 0$. 3
- ii) Hence solve $\sqrt{3} \cos x + \sin x = -1$ for $0 \leq x \leq 2\pi$ 2

Question 9 (20 marks)

a) For $(1+x)^{12}$, find the coefficient of x^8 . 1

b) For $(1+2x)^6$, find the term in x^4 . 1

c) i) Expand $(1+2x)^5$ as far as the term x^3 . 2

ii) Hence, find the coefficient of x^3 in the expansion of $(2-3x)(1+2x)^5$. 2

d) i) Find the general term in the expansion of $(2x^2 - x^{-1})^{20}$. 3

ii) Hence, find the term in x^{-5} . 2

e) i) Find the coefficients of x and x^{-3} in the expansion of $\left(3x - \frac{a}{x}\right)^5$. 2

ii) Hence, find the values of a if these coefficients are in the ratio 2:1 2

f) i) Use the binomial expansion

$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$ and by an appropriate substitution, prove that :

$$\binom{n}{0} + 2 \times \binom{n}{1} + 4 \times \binom{n}{2} + 8 \times \binom{n}{3} + \dots = 3^n \quad \text{2}$$

ii) Differentiate the identity $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$, then make

a substitution, to prove that

$$1 \times \binom{n}{1} + 2 \times \binom{n}{2} + \dots + n \times \binom{n}{n} = n2^{n-1} \quad \text{3}$$

Question 10 (12 marks)

- a) Prove by the method of mathematical induction that for all positive integer values of n :

$$5 + 11 + 19 + \dots + (8n - 3) = n(4n + 1) \quad 4$$

- b) Prove by mathematical induction that $2^{n+2} + 3^{2n+1}$ is divisible by 7 for all positive integers n . 4

- c) Prove by mathematical induction that $2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$ for all positive integers $n \geq 1$. 4

End of examination!!!

multiple choice questions.

$$\textcircled{1} \quad \frac{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}}{\cot \frac{\theta}{2} + \tan \frac{\theta}{2}} \quad \text{let } t = \tan \frac{\theta}{2}$$

$$= \frac{\frac{1}{t} - t}{\frac{1}{t} + t} = \frac{1 - t^2}{1 + t^2} = \cos \theta. \quad \textcircled{A}$$

$$\textcircled{2} \quad \cos 5x + \cos x = \frac{2}{2} [\cos(3x - 2x) + \cos(3x + 2x)]$$

$$\therefore \cos 5x + \cos x = 2 \cos 3x \cos 2x \quad \textcircled{D}$$

$$\textcircled{3} \quad \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{4 - \frac{3}{5}}{1 + 4 \times \frac{3}{5}} = \frac{17}{5} \times \frac{1}{1} = \frac{17}{5}$$

$$\therefore A - B = 45^\circ$$

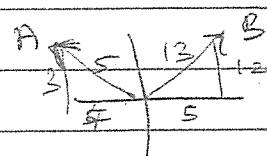
$\textcircled{A}$

$$\textcircled{4} \quad \sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$= \frac{3}{5} \times \frac{5}{13} - \left(-\frac{4}{5}\right) \times \frac{12}{13}$$

$$= \frac{15}{65} + \frac{48}{65} = \frac{63}{65}$$

$\textcircled{B}$



$$\sin A = \frac{3}{5}$$

$$\sin B = \frac{12}{13}$$

$$\textcircled{5} \quad (2 - 3x)^5 \Rightarrow \text{coefficient of } x^3 \text{ is}$$

$$8 \binom{5}{3} (-3)^3 =$$

$$= -48384$$

$\textcircled{D}$

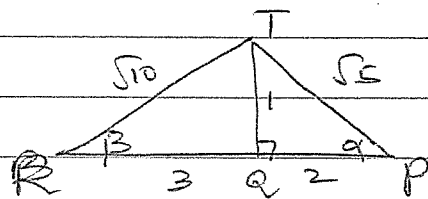
Question 6

a) (i) $\sin(A+B) \cdot \sin(A-B) = \sin^2 A - \sin^2 B$

LHS $\sin(A+B) \cdot \sin(A-B) = (\sin A \cos B + \cos A \sin B)(\sin A \cos B - \cos A \sin B)$
 $= \sin^2 A \cos^2 B + \sin A \cos A \cos A \sin B - \sin A \cos B \cdot \cos A \sin A - \cos^2 A \sin^2 B$
 $= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B$
 $= \sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B$
 $= \sin^2 A - \sin^2 A \sin^2 B - \sin^2 B + \sin^2 A \sin^2 B$
 $= \sin^2 A - \sin^2 B$

(ii) $\sin^2 75^\circ - \sin^2 15^\circ = \sin(75^\circ + 15^\circ) \cdot \sin(75^\circ - 15^\circ)$
 $= \sin 90^\circ \cdot \sin 60^\circ = \frac{\sqrt{3}}{2}$

b) (i)



$$\sin \alpha = \frac{1}{\sqrt{5}} \quad \text{and} \quad \sin \beta = \frac{1}{\sqrt{10}}$$

$$\begin{aligned} \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ &= \frac{1}{\sqrt{5}} \times \frac{3}{\sqrt{10}} + \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{10}} = \frac{3}{\sqrt{50}} + \frac{2}{\sqrt{50}} \\ &= \frac{5}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}} \end{aligned}$$

$$\sin(\alpha + \beta) = \frac{1}{\sqrt{2}}$$

(ii) $\therefore \alpha + \beta = 45^\circ$

c) (i) $\frac{\tan 110^\circ + \tan 25^\circ}{1 - \tan 110^\circ \tan 25^\circ} = \tan 135^\circ = -1$

(ii) $\sin \frac{8\pi}{9} \cdot \cos \frac{2\pi}{9} - \cos \frac{8\pi}{9} \cdot \sin \frac{2\pi}{9} = \sin\left(\frac{8\pi}{9} - \frac{2\pi}{9}\right) = \sin \frac{6\pi}{9}$
 $= \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

Question 7

a) $t = \tan \frac{\theta}{2}$

$$\therefore \frac{1 - \tan^2 \frac{3\pi}{8}}{1 + \tan^2 \frac{3\pi}{8}} = \frac{\cos \frac{6\pi}{8}}{\cos \frac{3\pi}{4}} = \frac{\cos \frac{3\pi}{4}}{\frac{1}{\sqrt{2}}} = -\frac{1}{\sqrt{2}}$$

(2)

b) $\cos \theta (\tan \theta - \tan \frac{\theta}{2}) = \tan \frac{\theta}{2}$

Let $t = \tan \frac{\theta}{2}$

LHS $\cos \theta (\tan \theta - \tan \frac{\theta}{2}) = \left(\frac{1-t^2}{1+t^2} \right) \left(\frac{2t}{1-t^2} - t \right)$ (3)

$$= \frac{1-t^2}{1+t^2} \times \left(\frac{2t-t+t^3}{1-t^2} \right)$$

$$= \frac{\cancel{1-t^2}}{1+t^2} \times \frac{t(\cancel{1-t^2}+t^2)}{\cancel{1-t^2}} = t$$

$$= \tan \frac{\theta}{2} = \text{RHS}$$

c) (i) $t = \tan 75^\circ$

$$\tan 150^\circ = \frac{2 \tan 75^\circ}{1 - \tan^2 75^\circ} = \frac{2t}{1-t^2} = -\frac{1}{\sqrt{3}} \quad (2)$$

(ii) $2\sqrt{3}t = -1 + t^2 \quad \therefore t^2 - 2\sqrt{3}t - 1 = 0$

$$\therefore t = \frac{2\sqrt{3} \pm \sqrt{12+4}}{2}$$

$$= \sqrt{3} \pm 2$$

(1)

$$\therefore \tan 75^\circ = 2 + \sqrt{3} \quad (75^\circ \text{ is acute } \angle)$$

d) Let $t = \tan \frac{\alpha}{2}$. Substituting $t = \tan \frac{\alpha}{2}$ gives

$$\frac{14t}{1+t^2} - \frac{4-4t^2}{1+t^2} = 4, \quad \alpha \neq 180^\circ.$$

$$\therefore 14t - 4 + 4t^2 = 4 + 4t^2 \Rightarrow 14t = 8 \quad \therefore t = \frac{4}{7}$$

$$\therefore \tan \frac{\alpha}{2} = \frac{4}{7} \Rightarrow \alpha = 59.29^\circ$$

and check for $x = 180^\circ$.

$$\underline{\text{LHS}} = 7 \times 0 - 4 \times (-1) = 4 = \text{RHS}$$

$\therefore x = 180^\circ$ is also a solution.

$$\therefore x = 180^\circ \text{ or } x = 59^\circ 29'$$

Questions

$$(i) \text{ LHS } \cos 36^\circ - \cos 84^\circ = \cos(A-B) - \cos(A+B)$$

$$A-B = 36 \text{ and}$$

$$A+B = 84$$

$$\therefore A = 60 \text{ and } B = 24$$

$$\begin{aligned} \therefore \cos 36^\circ - \cos 84^\circ &= 2 \sin 60^\circ \sin 24^\circ \\ &= \sqrt{3} \sin 24^\circ = \text{RHS} \end{aligned}$$

(2)

$$(ii) \frac{\sin 3\theta + \sin \theta}{\cos 3\theta + \cos \theta} = \tan 2\theta$$

$$\begin{aligned} \text{LHS } \frac{\sin 3\theta + \sin \theta}{\cos 3\theta + \cos \theta} &= \frac{2 \sin 2\theta \cdot \cos \theta}{2 \cos 2\theta \cdot \cos \theta} = \tan 2\theta \\ &= \text{RHS} \end{aligned}$$

$$b) \sin 2x + \sqrt{3} \cos x = 0$$

$$2 \sin x \cos x + \sqrt{3} \cos x = 0$$

$$\cos x [2 \sin x + \sqrt{3}] = 0$$

$$\therefore \cos x = 0 \text{ or } 2 \sin x = -\sqrt{3}$$

$$\sin x = -\frac{\sqrt{3}}{2}$$

$$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$x = \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}$$

c) (i)

$$\sqrt{3} \cos x + \sin x = A \cos(x - \theta)$$

Expanding $\sqrt{3} \cos x + \sin x = A \cos x \cos \theta + A \sin x \sin \theta$

equating
coefficients.

$$\sqrt{3} = A \cos \theta$$

$$1 = A \sin \theta$$

squaring
and adding,

$$\therefore 4 = A^2 \quad \therefore A = 2 \quad (A > 0)$$

$$\cos \theta = \frac{\sqrt{3}}{2} \quad \text{and} \quad \sin \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{6} \quad (\theta \text{ in Quadrant I})$$

$$\therefore \sqrt{3} \cos x + \sin x = 2 \cos \left(x - \frac{\pi}{6} \right)$$

(ii)

$$\sqrt{3} \cos x + \sin x = 2 \cos \left(x - \frac{\pi}{6} \right) = -1, \quad \text{or } x =$$

$$2 \cos \left(x - \frac{\pi}{6} \right) = -1$$

$$\therefore \cos \left(x - \frac{\pi}{6} \right) = -\frac{1}{2}$$

$$x - \frac{\pi}{6} = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\therefore x = \frac{2\pi}{3} + \frac{\pi}{6} \quad \text{or} \quad \frac{4\pi}{3} + \frac{\pi}{6}$$

$$= \frac{5\pi}{6} \quad \text{or} \quad \frac{9\pi}{6} = \frac{3\pi}{2}$$

Question 9

a) (i) $(1+x)^{12}$ coefficient of $x^8 = {}^{12}C_8$ (1)

(ii) $(1+2x)^6$ term in $x^4 = {}^6C_4 (2x)^4$
 $= 16 \times {}^6C_4 x^4$ (1)

c) (i) $(2-3x)(1+2x)^5 = (2-3x)(\dots + 10x + 40x^2 + 80x^3 + \dots)$ (2)

(ii) $\therefore$ coefficient of $x^3 = -120 + 160 = 40$ (2)

d) (i) $(2x^2 - x^{-1})^{20}$

General term = ${}^{20}C_r (2x^2)^{20-r} (-1)^r (x^{-1})^r$

$= (-1)^r \cdot {}^{20}C_r \cdot 2^{20-r} x^{40-3r}$ (3)

(ii) $40 - 3r = -5$

$\therefore 3r = 45 \quad \therefore r = 15$

$\therefore$ term in $x^{-5} = {}^{20}C_{15} (2x^2)^5 (-1)^{15} (x^{-1})^{15}$

$= -{}^{20}C_{15} \cdot 2^5 x^{-5}$

$$e) \left(3x - \frac{a}{x}\right)^5$$

$$\text{term of } x = {}^5C_2 \cdot (3x)^3 \left(-\frac{a}{x}\right)^2 \\ = {}^5C_2 \cdot 3^3 a^2$$

$$\text{term of } x^3 = {}^5C_4 \cdot (3x)^1 \left(-\frac{a}{x}\right)^4 \\ = {}^5C_4 \cdot 3 \cdot a^4$$

$$\frac{{}^5C_2 \cdot 3^3 a^2}{{}^5C_4 \cdot 3 \cdot a^4} = \frac{2}{1} \Rightarrow \frac{10}{5} \cdot \frac{3^2}{a^2} = \frac{2}{1}$$

$$\therefore a^2 = 9$$

$$\therefore a = \pm 3$$

$$f) (i) (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

Sub. in $x=2$

$$\text{LHS } 3^n \text{ and RHS } = {}^nC_0 + 2^n {}^nC_1 + 4^n {}^nC_2 + \dots + \dots$$

$$\therefore 3^n = {}^nC_0 + 2x {}^nC_1 + 4x {}^nC_2 + \dots \quad (2)$$

$$(ii) (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

$$\text{differentiating both sides } n(1+x)^{n-1} = {}^nC_1 + 2x {}^nC_2 + \dots + n {}^nC_n x^{n-1}$$

$$\text{sub in } x=1, n \cdot 2^{n-1} = {}^nC_1 + 2x {}^nC_2 + \dots + n {}^nC_n$$

(2)

Question 10

a) $5 + 11 + 19 + \dots + (8n-3) = n(4n+1)$ for all n

Prove true for $n=1$

$$\text{LHS } 5 \quad \text{RHS } = 1(5) = 5$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\therefore 5 + 11 + 19 + \dots + (8k-3) = k(4k+1) \quad \text{--- (1)}$$

Prove true for $n=k+1$

$$\text{ie } 5 + 11 + 19 + \dots + (8k-3) + (8k+5) = (k+1)(4k+5)$$

$$\text{LHS } 5 + 11 + 19 + \dots + (8k-3) + (8k+5)$$

=

$$= k(4k+1) + 8k+5 \quad \text{from (1)}$$

$$= 4k^2 + 9k + 5 = (k+1)(4k+5) = \text{RHS}$$

$\therefore$ true for $n=k+1$

$\therefore$ By the principle of m.I, it is true for $n=2, 3, \dots$ all positive integers of n

b) $2^{n+2} + 3^{2n+1}$ is divisible by 7

Prove true for $n=1$

$$2^3 + 3^3 = 35 \text{ is divisible by } 7$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$2^{k+2} + 3^{2k+1} = 7P \quad (P = \text{integer})$$

①

Prove true for $n = k+1$

$$2^{k+3} + 3^{2k+3} = 7Q \quad (Q = \text{integer})$$

LHS

$$2^{k+3} + 3^{2k+3} = 2^{k+2} + 9 \times 3^{2k+1}$$

$$= 2^{k+2} + 9(7P - 2^{k+2})$$

$$= 2 \cdot 2^{k+2} + 9 \times 7P - 9 \times 2^{k+2}$$

$$= 9 \times 7P - 7 \times 2^{k+2}$$

$$= 7[9P - 2^{k+2}] = 7Q$$

$\therefore$ By the principle of m.I., it is true for $n=1, 2, \dots$ all positive integers of n .

c)

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$$

Prove true for $n=1$

$$\underline{\text{LHS}} \quad 2 \times 1! = 2 \quad \underline{\text{RHS}} = 1(2!) = 2$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k! = k(k+1)! \quad \text{--- (1)}$$

Prove true for $n=k+1$

$$2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! + [(k+1)^2 + 1](k+1)! \\ = (k+1)(k+2)!$$

$$\underline{\text{LHS}} \quad 2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! + [(k+1)^2 + 1](k+1)!$$

$$= k(k+1)! + (k^2 + 2k + 2)(k+1)! \quad \text{from (1)}$$

$$= (k+1)! [k + k^2 + 2k + 2]$$

$$= (k+1)! [k^2 + 3k + 2] = (k+1)! (k+2)(k+1)$$

$$= (k+1)(k+2)! = \text{RHS}$$

By the principle of m.I, it is true
for $n=1, 2, \dots, n$.

